Signal Processing Practice Questions

1. Find Fourier Series Coefficients $a_0, a_1, and\ b_1$ for
$$x(t) = 3\cos(2\pi t)$$
Qualitatively sketch the signal. Is it periodic?

2. Find Complex Fourier Series coefficients, X_k for:
$$x(t) = 2\cos(4\pi t)$$
How many X_k will I have as non-zero values?

3. Find the Complex Fourier Series Coefficients, X_0, X_1, X_{-1} for:
$$x(t) = \begin{cases} t, & 0 \leq t \leq 1 \\ 1 - t, & 1 \leq t \leq 2 \end{cases}$$

4. Find the Fourier Transform of the following function:
$$x(t) = \exp(-a|t|)$$
Qualitatively sketch the FT.

5. Find the Fourier Transform of the following function
$$x(t) = \cos(6\pi t)$$

6. Find the Fourier Transform of the following function:
$$x(t) = \delta(t - 2)$$
Sketch the qualitatively Fourier Transform magnitude and phase.

7. Find the Discrete Time Fourier Transform of following signal. The sampling frequency is 1 Hz.
$$x[n] = \{1,2,1\}$$

8. Find the Discrete Fourier Transform of the following signal. The sampling frequency is 1 Hz.
$$x[n] = [1,0,-1,0]$$

9. Find the Discrete Fourier Transform of the following signal. The sampling frequency is 1 Hz.
$$x[n] = [1,1,1,1]$$

10. Use the FFT function in Matlab or Python and find Fourier Transform of $x(t) = \cos(6\pi t)$, by using a sampled signal with following sampling frequency:
(a) f_s= 4 Hz
(b) f_s= 100 Hz
Visualize these signal discretization in time domain and in frequency domain. What are the differences in time domain and in frequency domain? Compare it with what you should get analytically as calculated in Question (6).

11. Use the FFT function in Matlab or Python and find Fourier Transform of $x(t) = \exp(-a|t|)$, by using a sampled signal. Try out varying sampling frequencies. Compare the FFT with what you derived analytically in Question (5). What is the reason for discrepancy? Is there any way to avoid it?